# KAYCEE NDUKUBA

London, UK · kndukuba17@gmail.com
kndukuba17-hub.github.io · linkedin.com/in/kayceendukuba · github.com/kndukuba17-hub

## PROFESSIONAL SUMMARY

Applied Data Scientist (MSc) who pairs a psychology background with machine learning to model human behaviour and drive measurable business decisions. Built a portfolio of applied machine learning and analytics projects on real-world datasets spanning customer analytics, behavioural modelling, NLP and retrieval-augmented AI, with a focus on translating technical outputs into decisions people can act on — from a lifetime-value model that concentrates ~63% of future revenue in its top predicted decile to churn models reaching half of all churners in the riskiest 20%. Strong in Python, SQL and the scikit-learn / XGBoost stack, with an emphasis on model interpretability (SHAP).

## CORE SKILLS

**Programming & Databases:** Python (pandas, NumPy, scikit-learn), SQL (PostgreSQL, BigQuery, Snowflake), R, Git/GitHub
**Machine Learning & Statistics:** Predictive modelling, classification & regression, feature engineering, cross-validation, class-imbalance handling, model evaluation (ROC-AUC, precision/recall/F1), hypothesis testing, EDA
**Interpretability:** SHAP, feature importance, coefficient analysis
**NLP & AI:** Sentiment analysis, text classification, LLMs, Retrieval-Augmented Generation (LangChain, FAISS, sentence embeddings), prompt engineering
**Data Engineering & MLOps:** ETL, dbt, automation, data quality, Docker, GitHub Actions, MLflow, AWS, GCP
**Visualisation:** Matplotlib, Seaborn, Jupyter, dashboards & stakeholder reporting

## EXPERIENCE

**AI & Data Scientist** — Yves OS, London *Dec 2025 – Present*

- Developed behavioural analytics workflows that transformed multi-platform user-interaction data into measurable insights, used to prioritise product features and improve user experience.
- Applied large language models to generate personalised recommendations and summaries, then **evaluated outputs against defined quality criteria** to refine prompt design and improve analytical quality.
- Iteratively tested and refined AI-driven recommendation workflows using user feedback and exploratory analysis, improving recommendation relevance.
- Built engagement and productivity metrics that quantified user behaviour and informed data-driven product decisions, communicated to stakeholders through dashboards and reports.

*Python · SQL · Machine Learning · LLMs · pandas · PostgreSQL · AWS*

**Machine Learning Engineer** — Freelance / Contract *Jan 2025 – Dec 2025*

- Delivered applied machine learning and analytics across multiple private-sector engagements, translating business questions into models and measurable recommendations.
- **Selected engagements —** customer analytics, churn modelling, predictive modelling, demand forecasting and behavioural analysis.
- Built end-to-end analytical workflows — data preparation, feature engineering, model training and evaluation — in Python and SQL.
- Deployed and maintained production models with Docker on AWS/GCP, automating testing and model versioning via GitHub Actions and MLflow.

*Python · scikit-learn · SQL · AWS · GCP · Docker · GitHub Actions · MLflow*

**Data Analyst** — Plan-Net, Leicester *Oct 2023 – Nov 2024*

- Automated recurring reporting with SQL and Python, reducing manual effort and improving turnaround for business stakeholders.
- Designed and maintained ETL workflows that improved data quality, consistency and reliability across reporting pipelines.
- Partnered with cross-functional stakeholders to translate operational data into clear reports and recommendations that supported decision-making.
- Analysed infrastructure and operational data to surface trends, recurring issues and opportunities for optimisation.

*SQL · Python · ETL · dbt · Snowflake · Git*

## SELECTED PROJECTS (FULL CODE: GITHUB.COM/KNDUKUBA17-HUB)

**Customer Lifetime Value (CLV) Prediction** · *Python, XGBoost, SHAP*

- Modelled 6-month customer spend from RFM behaviour on **1.07M real transactions** (UCI Online Retail II) using a leakage-safe temporal train/test split.

- XGBoost reached test $R^2$ 0.69 / MAE £600; the **top-10% predicted customers captured ~63% of actual future revenue** (vs 10% at random) — a ready-to-use targeting list.
- Ranked the behavioural drivers of value with SHAP to make the output actionable for marketing.

**Customer Churn Prediction** · *Python, scikit-learn, XGBoost, SHAP*

- Predicted churn on **7,043 real telecom customers**; Logistic Regression achieved **ROC-AUC 0.842** (5-fold CV 0.845) with 0.78 recall on the churn class.
- Optimised for recall/ROC-AUC over accuracy (73.5% majority baseline); targeting the top-20% risk reaches **~50% of churners**.
- Handled class imbalance and explained drivers with coefficients and SHAP.

**Big Five Personality Analysis (Psychology × Data Science)** · *Python, PCA, K-Means*

- Validated the Big Five factor structure on **603k real survey responses** (within-trait item correlation 0.37 vs 0.01 across traits; PCA elbow at five).
- Built behavioural segmentation into four personality types with K-Means, with an honest read that personality is dimensional rather than categorical.

**Enterprise RAG Knowledge Assistant** · *Python, LangChain, FAISS*

- Built the retrieval layer of a Retrieval-Augmented Generation system: document chunking, sentence-transformer embeddings and FAISS semantic search over policy documents.

**Aviation Sentiment Classifier (NLP)** · *Python, scikit-learn, gensim*

- Classified **14,640 real airline tweets**; Bag-of-Words + Logistic Regression (GridSearchCV) reached **80.4% accuracy / 0.74 macro-F1**, outperforming Word2Vec, using custom scikit-learn transformers.

**Customer Churn Dashboard (interactive BI)** · *JavaScript, SVG, HTML*

- Built an interactive retention dashboard on **7,043 real customers** with live client-side cross-filtering, KPI tiles (churn rate, monthly revenue at risk) and hand-built charts — no libraries. Live: kndukuba17-hub.github.io/Customer-Churn-Dashboard

## EDUCATION

**MSc, Applied Data Science & Cyber Security** — Royal Holloway, University of London *2025 – 2026*
Modules: Machine Learning, Neural Networks, Big Data Analytics, Statistical Modelling, Cyber Threat Intelligence.

**BSc, Psychology** — University of Leicester *2021 – 2024*
Modules: Quantitative Research Methods, Behavioural Statistics, Cognitive Neuroscience.

**A-Levels** — Langley Park School *2019 – 2021*
Modules: Chemistry (A), Biology (A), Psychology (A).